

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning, 3rd Year

Experience

Bentham AI

Dec 2025 – Feb 2026

AI Engineering Intern: Python, LangChain, LangGraph, Go, Docker, GCP, GitHub Actions

Remote

- Built **agentic AI workflows** for **legal document drafting and validation** using **Python, LangChain/LangGraph**, Docker, and GCP, reducing total processing time by **88%**.
- Implemented **secure document processing pipelines** with encrypted storage and role-based access control, supporting **100+ customer submissions** and **500+ document uploads**.
- Built **monitoring and logging** for **production AI services** to surface failed AI requests, trace workflow failures, and cut debugging time by **30%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalius: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built a **brain subpart segmentation** tool using **FastSurfer CNN models** to segment **95 anatomical brain regions** from **MRI volumes**, including preprocessing, image conformation, inference orchestration, and post-processing.
- Converted models to **ONNX** and added **PyTorch** and **TinyGrad inference** paths, then generated **label-mapped segmentation volumes** and **binary masks** for cortical/subcortical structures.
- Reduced expensive processing runtime by **87%** through **async/parallel execution** while debugging GPU/CPU paths, large MRI data movement, generated masks, and output correctness.

Open Source

Contributions

- Contributed maintainer-reviewed patches across **Kuadrant MCP Gateway, Kyverno, Invesalius, CRIU, and VideoLAN**, spanning backend reliability, security validation, parser robustness, compatibility fixes, and artifact correctness.

Projects

Hyoka Agent Reliability Layer | *Python, FastAPI, SQLAlchemy, PostgreSQL, OpenTelemetry, Next.js*

[GitHub](#)

- Built an **AI evaluation layer** that ingests **agent traces**, mines recurring failures, replays faulty runs, and generates **candidate fixes** for safer agent behavior.
- Implemented an **OpenAI-compatible proxy**, **OTLP/OpenInference-style ingestion**, **Python SDK hooks**, worker leases, artifacts, manifests, release gates, and a Next.js dashboard for trace/run/gate inspection.
- Designed SQLite/PostgreSQL metadata models so agent changes could be **reproduced, compared, promoted**, or **rolled back** through explicit evaluation gates.

Swish Support Automation | *Python, FastAPI, React, PostgreSQL, Redis, Langfuse*

[GitHub](#)

- Built a **customer support agent** to replace **templated replies** with **context-aware responses** grounded in order, kitchen, fleet, trust, evidence, issue, and conversation state.
- Kept refunds, coupons, replacements, and escalations **policy-bound**; PostgreSQL/Redis stored support sessions, issue signals, customer context, and follow-up behavior.
- Added **Langfuse observability**, local traces, pytest regression suites, **response-quality checks**, and deterministic fallbacks so support behavior stayed inspectable.

Penny Lane Multi-Agent Workflow | *Python, Agno, Pydantic, SQLite, JSONL, React/Vite*

[GitHub](#)

- Built a **multi-agent trading-firm simulation** with analyst, bull/bear debate, research manager, trader, **risk-review**, portfolio-approval, and evaluation stages wired through typed state.
- Used **Pydantic state**, SQLite checkpoints, **JSONL/full-state traces**, baseline comparisons, validation harnesses, and a React/Vite inspection UI to review investment decisions and agent reasoning.

Technical Skills

Languages: Python, TypeScript, JavaScript, Go

Frameworks: LangChain, LangGraph, FastAPI, React, Next.js, SQLAlchemy, PyTorch, Pydantic

Tools: PostgreSQL, Redis, Docker, GCP, GitHub Actions, OpenTelemetry, Langfuse, ONNX, REST APIs